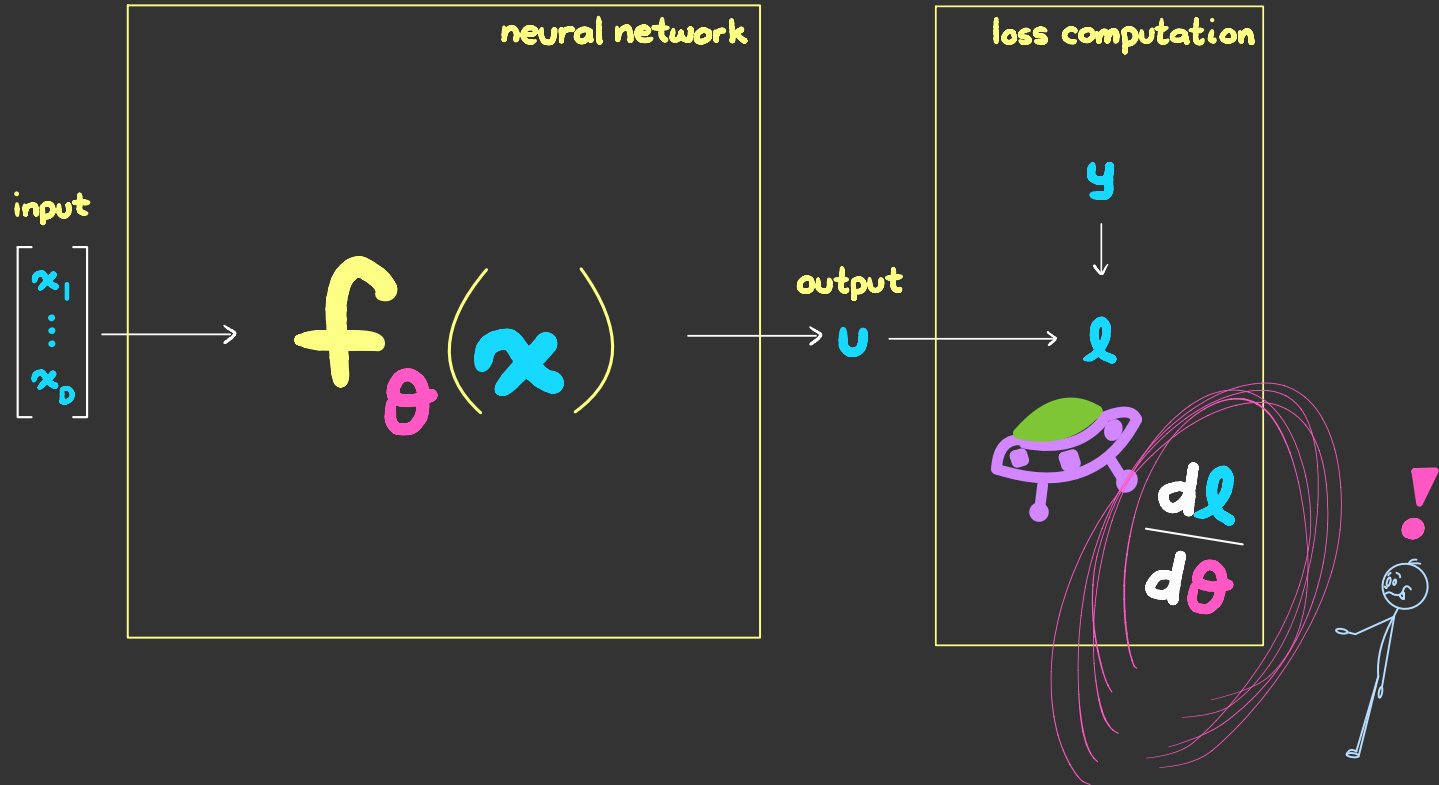


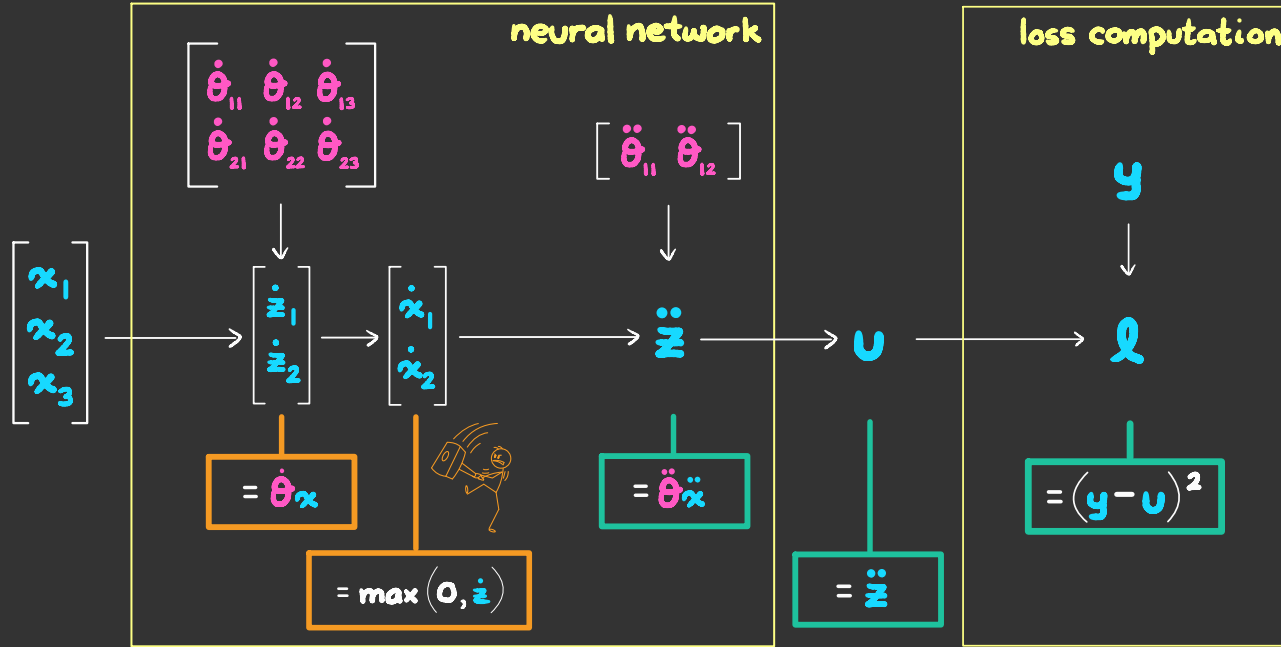
optimizing neural networks

CSCI
270

so how do we compute the loss gradient?



let's try it on a small example



$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}_{ij}} = ?$$

$$\frac{\partial \mathcal{L}}{\partial \ddot{\theta}_{ij}} = ?$$

$$\begin{bmatrix} \dot{\theta}_{11} & \dot{\theta}_{12} & \dot{\theta}_{13} \\ \dot{\theta}_{21} & \dot{\theta}_{22} & \dot{\theta}_{23} \end{bmatrix}$$

$$\begin{bmatrix} \ddot{\theta}_{11} & \ddot{\theta}_{12} \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix}$$

$$\ddot{x}$$

$$u$$

$$\mathcal{L}$$

$$y$$

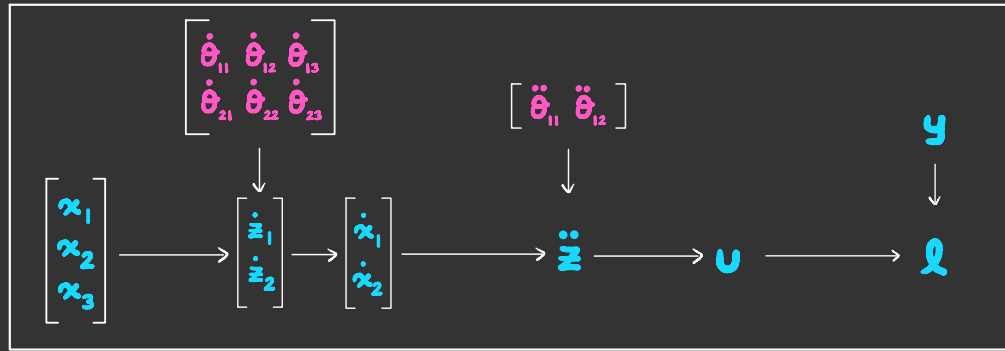
$$= \dot{\theta} x$$

$$= \max(0, \dot{x})$$

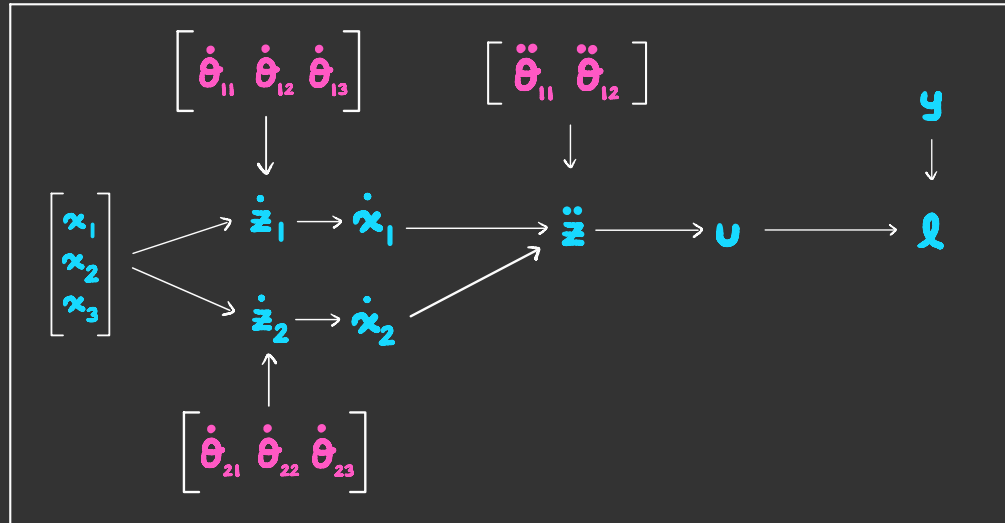
$$= \ddot{\theta} \ddot{x}$$

$$= \ddot{x}$$

$$= (y - u)^2$$



can be rewritten:



$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}_{1j}} = ?$$

$$\frac{\partial \mathcal{L}}{\partial \ddot{\theta}_{1j}} = ?$$

$$\begin{bmatrix} \dot{\theta}_{11} & \dot{\theta}_{12} & \dot{\theta}_{13} \end{bmatrix}$$

$$\begin{bmatrix} \ddot{\theta}_{11} & \ddot{\theta}_{12} \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\ddot{x}_1$$

$$\dot{x}_1$$

$$\ddot{x}$$

$$u$$

$$\mathcal{L}$$

$$\ddot{x}_2$$

$$\dot{x}_2$$

$$\begin{bmatrix} \dot{\theta}_{21} & \dot{\theta}_{22} & \dot{\theta}_{23} \end{bmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}_{2j}} = ?$$

$$y$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}_{1j}} = ?$$

$$\frac{\partial \mathcal{L}}{\partial \ddot{\theta}_{ij}} = \frac{\partial \mathcal{L}}{\partial u} \frac{\partial u}{\partial \ddot{\theta}_{ij}}$$

$$\begin{bmatrix} \dot{\theta}_{11} & \dot{\theta}_{12} & \dot{\theta}_{13} \end{bmatrix}$$

$$\begin{bmatrix} \ddot{\theta}_{11} & \ddot{\theta}_{12} \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\begin{aligned} \ddot{z}_1 &\rightarrow \dot{x}_1 \\ \ddot{z}_2 &\rightarrow \dot{x}_2 \end{aligned}$$

$$\begin{bmatrix} \dot{\theta}_{21} & \dot{\theta}_{22} & \dot{\theta}_{23} \end{bmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}_{2j}} = ?$$

$$\ddot{z}$$

$$u$$

$$y \rightarrow \mathcal{L}$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}_{1j}} = ?$$

$$\frac{\partial \mathcal{L}}{\partial \ddot{\theta}_{ij}} = \frac{\partial \mathcal{L}}{\partial u} \frac{\partial u}{\partial \ddot{x}} \frac{\partial \ddot{x}}{\partial \ddot{\theta}_{ij}}$$

$$\begin{bmatrix} \dot{\theta}_{11} & \dot{\theta}_{12} & \dot{\theta}_{13} \end{bmatrix}$$

$$\begin{bmatrix} \ddot{\theta}_{11} & \ddot{\theta}_{12} \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\begin{aligned} \ddot{x}_1 &\rightarrow \dot{x}_1 \\ \ddot{x}_2 &\rightarrow \dot{x}_2 \end{aligned}$$

$$\ddot{x}$$

$$u$$

$$y$$

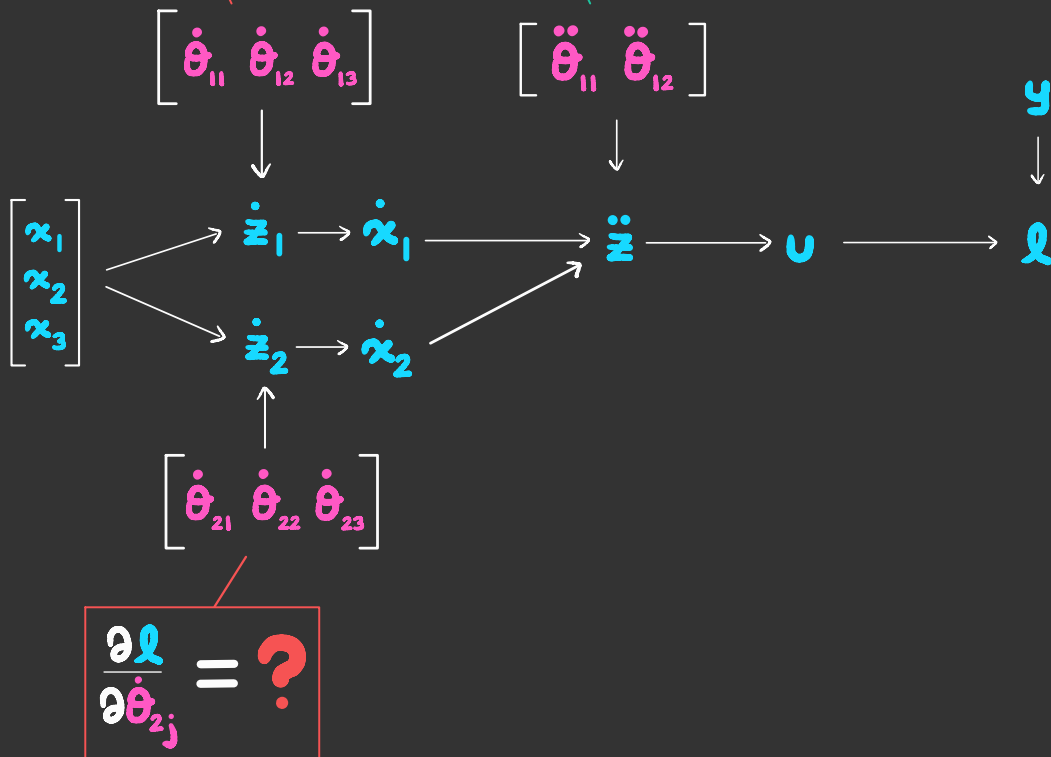
$$\mathcal{L}$$

$$\begin{bmatrix} \dot{\theta}_{21} & \dot{\theta}_{22} & \dot{\theta}_{23} \end{bmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}_{2j}} = ?$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}_{ij}} = ?$$

$$\frac{\partial \mathcal{L}}{\partial \ddot{\theta}_{ij}} = \frac{\partial \mathcal{L}}{\partial u} \frac{\partial u}{\partial \ddot{x}} \frac{\partial \ddot{x}}{\partial \ddot{\theta}_{ij}}$$



$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}_{ij}} = \frac{\partial \mathcal{L}}{\partial u} \frac{\partial u}{\partial \dot{\theta}_{ij}}$$

$$\frac{\partial \mathcal{L}}{\partial \ddot{\theta}_{ij}} = \frac{\partial \mathcal{L}}{\partial u} \frac{\partial u}{\partial \ddot{\theta}_{ij}} \frac{\partial \ddot{\theta}_{ij}}{\partial \ddot{\theta}_{ij}}$$

$$\begin{bmatrix} \dot{\theta}_{11} & \dot{\theta}_{12} & \dot{\theta}_{13} \end{bmatrix}$$

$$\begin{bmatrix} \ddot{\theta}_{11} & \ddot{\theta}_{12} \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\ddot{x}_1$$

$$\dot{x}_1$$

$$\ddot{x}$$

$$u$$

$$y$$

$$\mathcal{L}$$

$$\ddot{x}_2$$

$$\dot{x}_2$$

$$\begin{bmatrix} \dot{\theta}_{21} & \dot{\theta}_{22} & \dot{\theta}_{23} \end{bmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}_{2j}} = \frac{\partial \mathcal{L}}{\partial u} \frac{\partial u}{\partial \dot{\theta}_{2j}}$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}_{1j}} = \frac{\partial \mathcal{L}}{\partial u} \frac{\partial u}{\partial \ddot{\theta}_{1j}} \frac{\partial \ddot{\theta}_{1j}}{\partial \dot{\theta}_{1j}}$$

$$\frac{\partial \mathcal{L}}{\partial \ddot{\theta}_{1j}} = \frac{\partial \mathcal{L}}{\partial u} \frac{\partial u}{\partial \ddot{\theta}_{1j}} \frac{\partial \ddot{\theta}_{1j}}{\partial \ddot{\theta}_{1j}}$$

$$\begin{bmatrix} \dot{\theta}_{11} & \dot{\theta}_{12} & \dot{\theta}_{13} \end{bmatrix}$$

$$\begin{bmatrix} \ddot{\theta}_{11} & \ddot{\theta}_{12} \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\ddot{x}_1$$

$$\dot{x}_1$$

$$\ddot{x}$$

$$u$$

$$y$$

$$l$$

$$\ddot{x}_2$$

$$\dot{x}_2$$

$$\begin{bmatrix} \dot{\theta}_{21} & \dot{\theta}_{22} & \dot{\theta}_{23} \end{bmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}_{2j}} = \frac{\partial \mathcal{L}}{\partial u} \frac{\partial u}{\partial \ddot{\theta}_{2j}} \frac{\partial \ddot{\theta}_{2j}}{\partial \dot{\theta}_{2j}}$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}_{1j}} = \frac{\partial \mathcal{L}}{\partial \mathbf{u}} \frac{\partial \mathbf{u}}{\partial \ddot{\mathbf{x}}} \frac{\partial \ddot{\mathbf{x}}}{\partial \dot{\mathbf{x}}_1} \frac{\partial \dot{\mathbf{x}}_1}{\partial \dot{\theta}_{1j}}$$

$$\frac{\partial \mathcal{L}}{\partial \ddot{\theta}_{ij}} = \frac{\partial \mathcal{L}}{\partial \mathbf{u}} \frac{\partial \mathbf{u}}{\partial \ddot{\mathbf{x}}} \frac{\partial \ddot{\mathbf{x}}}{\partial \ddot{\theta}_{ij}}$$

$$\begin{bmatrix} \dot{\theta}_{11} & \dot{\theta}_{12} & \dot{\theta}_{13} \end{bmatrix}$$

$$\begin{bmatrix} \ddot{\theta}_{11} & \ddot{\theta}_{12} \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\dot{\mathbf{x}}_1$$

$$\dot{\mathbf{x}}_1$$

$$\ddot{\mathbf{x}}$$

$$\mathbf{u}$$

$$\mathbf{y}$$

$$\mathcal{L}$$

$$\dot{\mathbf{x}}_2$$

$$\dot{\mathbf{x}}_2$$

$$\begin{bmatrix} \dot{\theta}_{21} & \dot{\theta}_{22} & \dot{\theta}_{23} \end{bmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}_{2j}} = \frac{\partial \mathcal{L}}{\partial \mathbf{u}} \frac{\partial \mathbf{u}}{\partial \ddot{\mathbf{x}}} \frac{\partial \ddot{\mathbf{x}}}{\partial \dot{\mathbf{x}}_2} \frac{\partial \dot{\mathbf{x}}_2}{\partial \dot{\theta}_{2j}}$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}_{1j}} = \frac{\partial \mathcal{L}}{\partial u} \frac{\partial u}{\partial \ddot{x}} \frac{\partial \ddot{x}}{\partial \dot{x}_1} \frac{\partial \dot{x}_1}{\partial \dot{z}_1} \frac{\partial \dot{z}_1}{\partial \dot{\theta}_{1j}}$$

$$\frac{\partial \mathcal{L}}{\partial \ddot{\theta}_{ij}} = \frac{\partial \mathcal{L}}{\partial u} \frac{\partial u}{\partial \ddot{x}} \frac{\partial \ddot{x}}{\partial \ddot{\theta}_{ij}}$$

$$\begin{bmatrix} \dot{\theta}_{11} & \dot{\theta}_{12} & \dot{\theta}_{13} \end{bmatrix}$$

$$\begin{bmatrix} \ddot{\theta}_{11} & \ddot{\theta}_{12} \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\dot{z}_1$$

$$\dot{x}_1$$

$$\ddot{x}$$

$$u$$

$$\mathcal{L}$$

$$y$$

$$\dot{z}_2$$

$$\dot{x}_2$$

$$\begin{bmatrix} \dot{\theta}_{21} & \dot{\theta}_{22} & \dot{\theta}_{23} \end{bmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}_{2j}} = \frac{\partial \mathcal{L}}{\partial u} \frac{\partial u}{\partial \ddot{x}} \frac{\partial \ddot{x}}{\partial \dot{x}_2} \frac{\partial \dot{x}_2}{\partial \dot{z}_2} \frac{\partial \dot{z}_2}{\partial \dot{\theta}_{2j}}$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}_{1j}} = \frac{\partial \mathcal{L}}{\partial u} \frac{\partial u}{\partial \ddot{x}} \frac{\partial \ddot{x}}{\partial \dot{x}_1} \frac{\partial \dot{x}_1}{\partial \dot{\theta}_{1j}}$$

$$\frac{\partial \mathcal{L}}{\partial \ddot{\theta}_{ij}} = \frac{\partial \mathcal{L}}{\partial u} \frac{\partial u}{\partial \ddot{x}} \frac{\partial \ddot{x}}{\partial \ddot{\theta}_{ij}}$$

$$\begin{bmatrix} \dot{\theta}_{11} & \dot{\theta}_{12} & \dot{\theta}_{13} \end{bmatrix}$$

$$\begin{bmatrix} \ddot{\theta}_{11} & \ddot{\theta}_{12} \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\dot{x}_1 \rightarrow \ddot{x}_1$$

$$\dot{x}_2 \rightarrow \ddot{x}_2$$

$$\ddot{x}$$

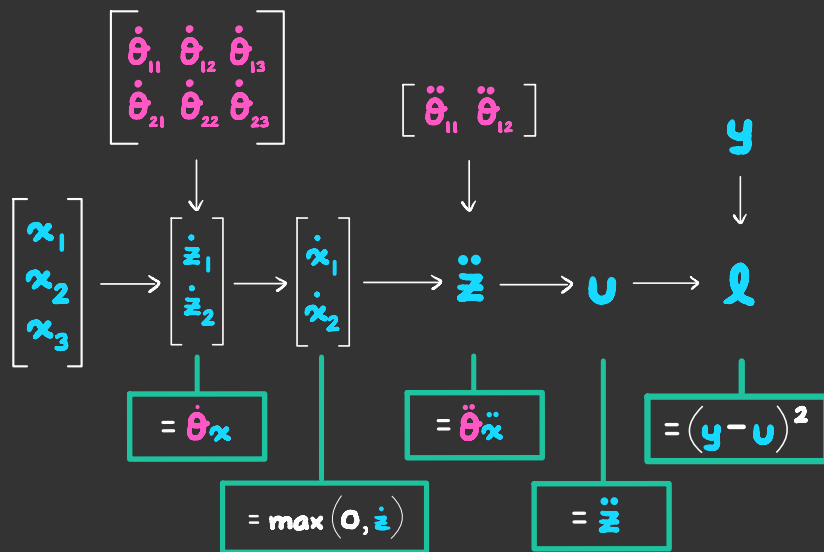
$$u$$

$$\mathcal{L}$$

$$y$$

$$\begin{bmatrix} \dot{\theta}_{21} & \dot{\theta}_{22} & \dot{\theta}_{23} \end{bmatrix}$$

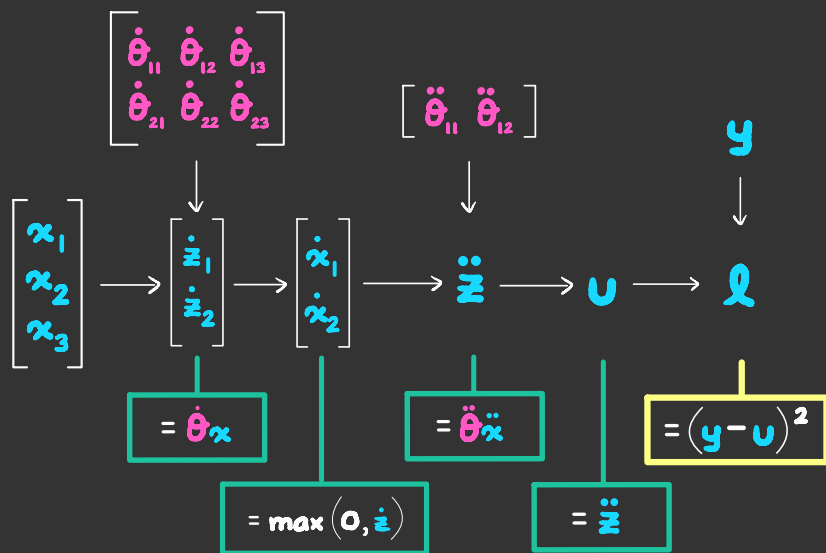
$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}_{2j}} = \frac{\partial \mathcal{L}}{\partial u} \frac{\partial u}{\partial \ddot{x}} \frac{\partial \ddot{x}}{\partial \dot{x}_2} \frac{\partial \dot{x}_2}{\partial \dot{\theta}_{2j}}$$



$$\frac{\partial l}{\partial \dot{\theta}_{1j}} = \frac{\partial l}{\partial u} \frac{\partial u}{\partial \ddot{z}} \frac{\partial \ddot{z}}{\partial \dot{x}_1} \frac{\partial \dot{x}_1}{\partial \dot{z}_1} \frac{\partial \dot{z}_1}{\partial \dot{\theta}_{1j}}$$

$$\frac{\partial l}{\partial \dot{\theta}_{2j}} = \frac{\partial l}{\partial u} \frac{\partial u}{\partial \ddot{z}} \frac{\partial \ddot{z}}{\partial \dot{x}_2} \frac{\partial \dot{x}_2}{\partial \dot{z}_2} \frac{\partial \dot{z}_2}{\partial \dot{\theta}_{2j}}$$

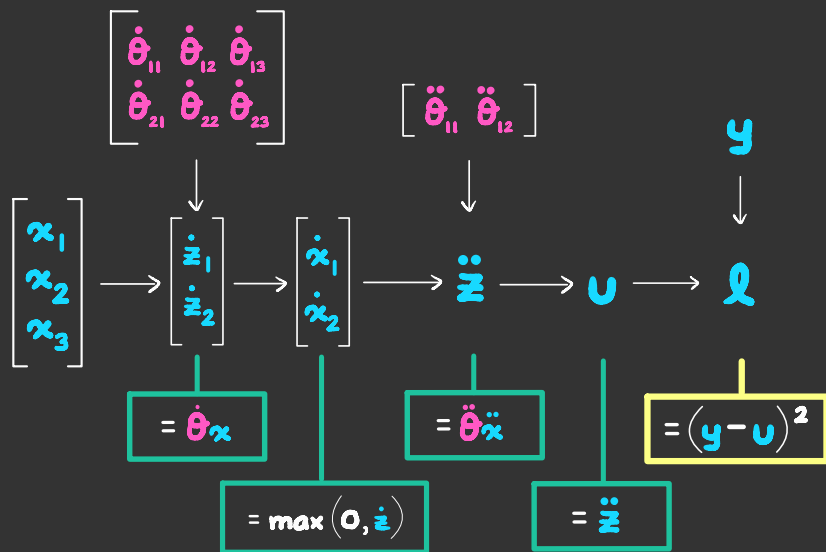
$$\frac{\partial l}{\partial \ddot{\theta}_{ij}} = \frac{\partial l}{\partial u} \frac{\partial u}{\partial \ddot{z}} \frac{\partial \ddot{z}}{\partial \ddot{\theta}_{ij}}$$



$$\frac{\partial l}{\partial \dot{\theta}_{1j}} = \boxed{\frac{\partial l}{\partial u}} \frac{\partial u}{\partial \ddot{x}} \frac{\partial \ddot{x}}{\partial \dot{x}_1} \frac{\partial \dot{x}_1}{\partial \dot{x}_1} \frac{\partial \dot{x}_1}{\partial \dot{\theta}_{1j}}$$

$$\frac{\partial l}{\partial \dot{\theta}_{2j}} = \boxed{\frac{\partial l}{\partial u}} \frac{\partial u}{\partial \ddot{x}} \frac{\partial \ddot{x}}{\partial \dot{x}_2} \frac{\partial \dot{x}_2}{\partial \dot{x}_2} \frac{\partial \dot{x}_2}{\partial \dot{\theta}_{2j}}$$

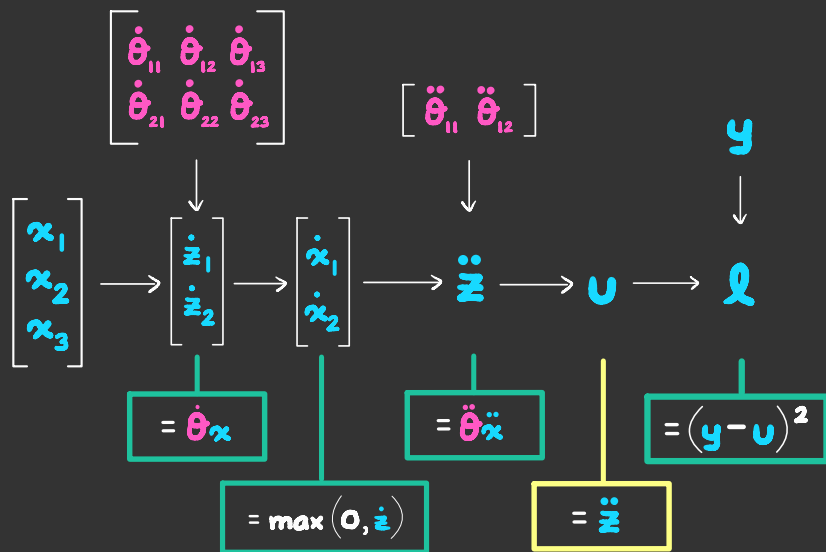
$$\frac{\partial l}{\partial \ddot{\theta}_{ij}} = \boxed{\frac{\partial l}{\partial u}} \frac{\partial u}{\partial \ddot{x}} \frac{\partial \ddot{x}}{\partial \ddot{\theta}_{ij}}$$



$$\frac{\partial l}{\partial \dot{\theta}_{1j}} = (-2(y-u)) \frac{\partial u}{\partial \ddot{z}} \frac{\partial \ddot{z}}{\partial \dot{x}_1} \frac{\partial \dot{x}_1}{\partial \dot{z}_1} \frac{\partial \dot{z}_1}{\partial \dot{\theta}_{1j}}$$

$$\frac{\partial l}{\partial \dot{\theta}_{2j}} = (-2(y-u)) \frac{\partial u}{\partial \ddot{z}} \frac{\partial \ddot{z}}{\partial \dot{x}_2} \frac{\partial \dot{x}_2}{\partial \dot{z}_2} \frac{\partial \dot{z}_2}{\partial \dot{\theta}_{2j}}$$

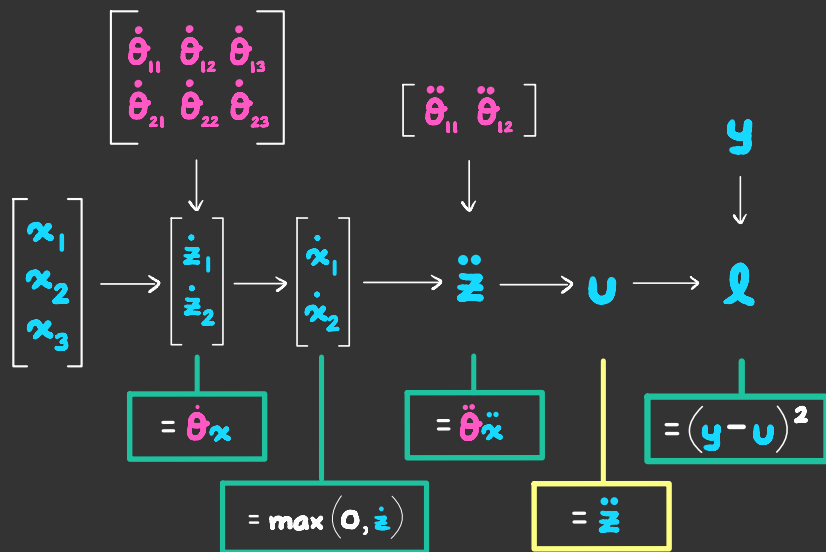
$$\frac{\partial l}{\partial \ddot{\theta}_{ij}} = (-2(y-u)) \frac{\partial u}{\partial \ddot{z}} \frac{\partial \ddot{z}}{\partial \ddot{\theta}_{ij}}$$



$$\frac{\partial l}{\partial \dot{\theta}_{1j}} = (-2(y-u)) \left[\frac{\partial u}{\partial \ddot{z}} \frac{\partial \ddot{z}}{\partial \dot{x}_1} \frac{\partial \dot{x}_1}{\partial \dot{z}_1} \frac{\partial \dot{z}_1}{\partial \dot{\theta}_{1j}} \right]$$

$$\frac{\partial l}{\partial \dot{\theta}_{2j}} = (-2(y-u)) \left[\frac{\partial u}{\partial \ddot{z}} \frac{\partial \ddot{z}}{\partial \dot{x}_2} \frac{\partial \dot{x}_2}{\partial \dot{z}_2} \frac{\partial \dot{z}_2}{\partial \dot{\theta}_{2j}} \right]$$

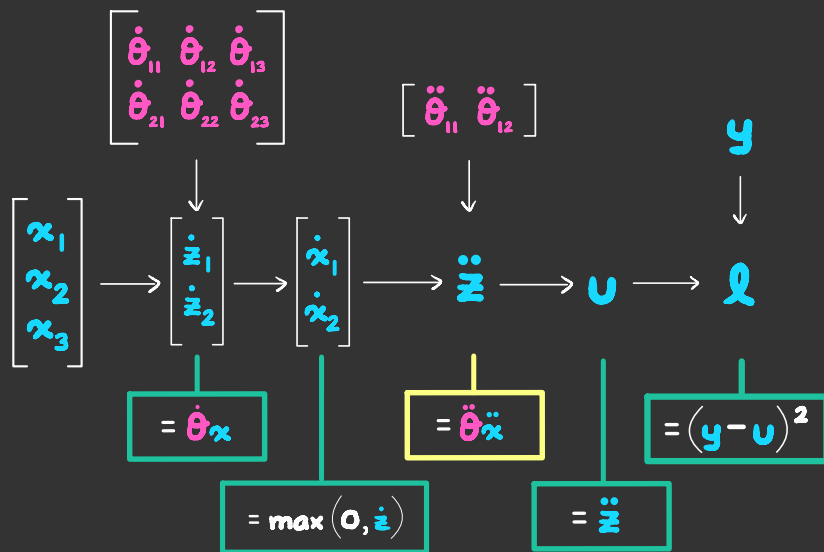
$$\frac{\partial l}{\partial \ddot{\theta}_{ij}} = (-2(y-u)) \left[\frac{\partial u}{\partial \ddot{z}} \frac{\partial \ddot{z}}{\partial \ddot{\theta}_{ij}} \right]$$



$$\frac{\partial l}{\partial \dot{\theta}_{1j}} = (-2(y-u)) \cdot \boxed{1} \cdot \frac{\partial \ddot{x}}{\partial \dot{x}_1} \frac{\partial \dot{x}_1}{\partial \dot{z}_1} \frac{\partial \dot{z}_1}{\partial \dot{\theta}_{1j}}$$

$$\frac{\partial l}{\partial \dot{\theta}_{2j}} = (-2(y-u)) \cdot \boxed{1} \cdot \frac{\partial \ddot{x}}{\partial \dot{x}_2} \frac{\partial \dot{x}_2}{\partial \dot{z}_2} \frac{\partial \dot{z}_2}{\partial \dot{\theta}_{2j}}$$

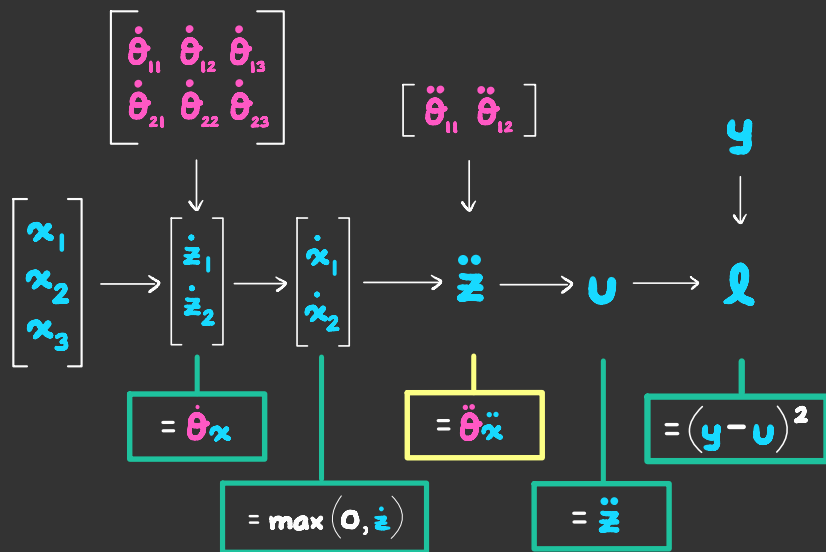
$$\frac{\partial l}{\partial \ddot{\theta}_{ij}} = (-2(y-u)) \cdot \boxed{1} \cdot \frac{\partial \ddot{x}}{\partial \ddot{\theta}_{ij}}$$



$$\frac{\partial l}{\partial \dot{\theta}_{ij}} = (-2(y-u)) \cdot 1 \cdot \frac{\partial \ddot{z}}{\partial \dot{x}_1} \frac{\partial \dot{x}_1}{\partial \dot{z}_1} \frac{\partial \dot{z}_1}{\partial \dot{\theta}_{ij}}$$

$$\frac{\partial l}{\partial \dot{\theta}_{2j}} = (-2(y-u)) \cdot 1 \cdot \frac{\partial \ddot{z}}{\partial \dot{x}_2} \frac{\partial \dot{x}_2}{\partial \dot{z}_2} \frac{\partial \dot{z}_2}{\partial \dot{\theta}_{2j}}$$

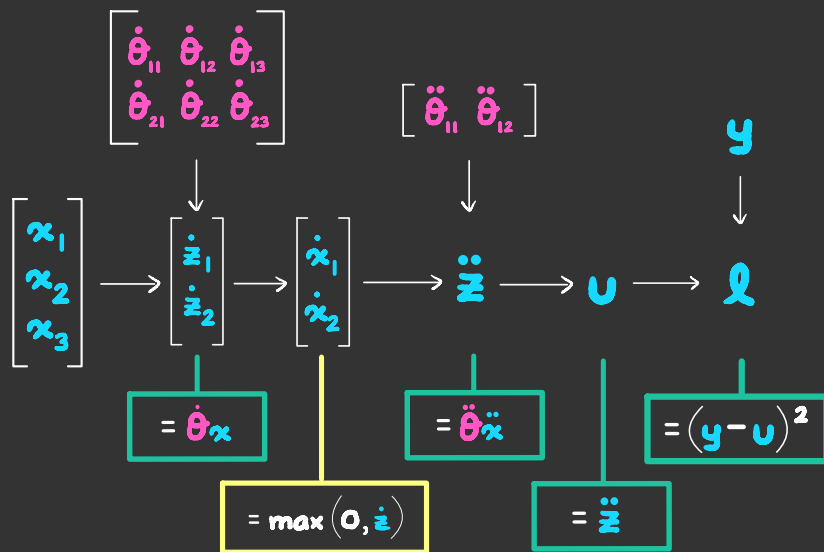
$$\frac{\partial l}{\partial \ddot{\theta}_{ij}} = (-2(y-u)) \cdot 1 \cdot \frac{\partial \ddot{z}}{\partial \ddot{\theta}_{ij}}$$



$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}_{1j}} = (-2(y-u)) \cdot 1 \cdot \ddot{\theta}_{11} \cdot \frac{\partial x_1}{\partial \dot{x}_1} \frac{\partial \dot{x}_1}{\partial \dot{\theta}_{1j}}$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}_{2j}} = (-2(y-u)) \cdot 1 \cdot \ddot{\theta}_{12} \cdot \frac{\partial x_2}{\partial \dot{x}_2} \frac{\partial \dot{x}_2}{\partial \dot{\theta}_{2j}}$$

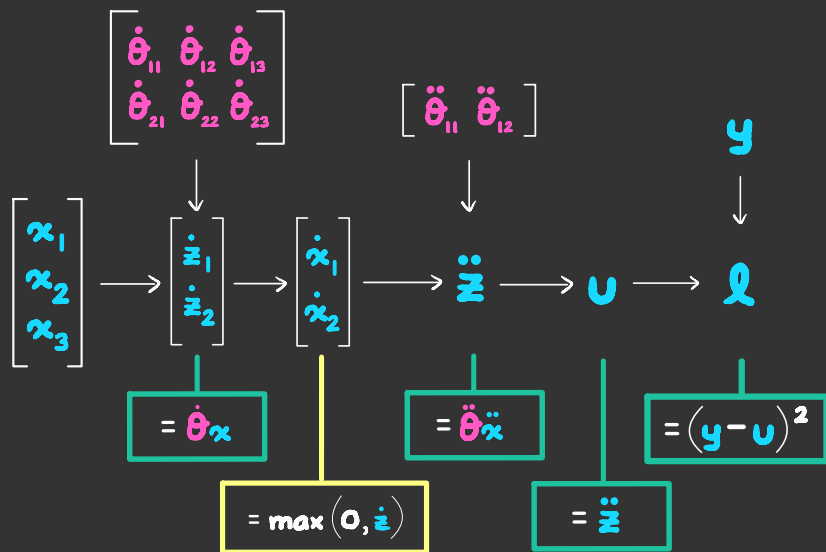
$$\frac{\partial \mathcal{L}}{\partial \ddot{\theta}_{ij}} = (-2(y-u)) \cdot 1 \cdot \frac{\partial \ddot{x}}{\partial \ddot{\theta}_{ij}}$$



$$\frac{\partial l}{\partial \dot{\theta}_{1j}} = (-2(y-u)) \cdot 1 \cdot \ddot{\theta}_{11} \cdot \frac{\partial \dot{z}_1}{\partial \dot{z}_1} \frac{\partial \dot{z}_1}{\partial \dot{\theta}_{1j}}$$

$$\frac{\partial l}{\partial \dot{\theta}_{2j}} = (-2(y-u)) \cdot 1 \cdot \ddot{\theta}_{12} \cdot \frac{\partial \dot{z}_2}{\partial \dot{z}_2} \frac{\partial \dot{z}_2}{\partial \dot{\theta}_{2j}}$$

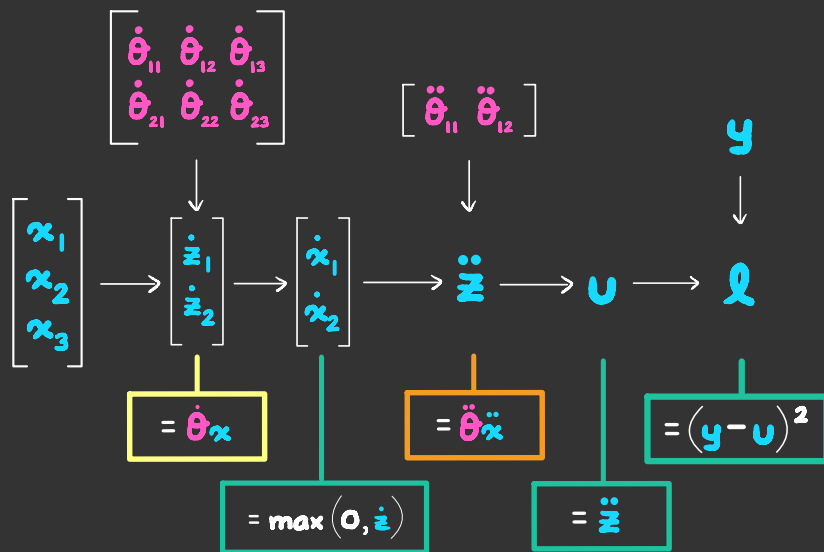
$$\frac{\partial l}{\partial \ddot{\theta}_{ij}} = (-2(y-u)) \cdot 1 \cdot \frac{\partial \ddot{z}}{\partial \ddot{\theta}_{ij}}$$



$$\frac{\partial l}{\partial \dot{\theta}_{1j}} = (-2(y-u)) \cdot 1 \cdot \ddot{\theta}_{11} \cdot \boxed{1(\dot{z}_1 > 0)} \cdot \frac{\partial \dot{z}_1}{\partial \dot{\theta}_{1j}}$$

$$\frac{\partial l}{\partial \dot{\theta}_{2j}} = (-2(y-u)) \cdot 1 \cdot \ddot{\theta}_{12} \cdot \boxed{1(\dot{z}_2 > 0)} \cdot \frac{\partial \dot{z}_2}{\partial \dot{\theta}_{2j}}$$

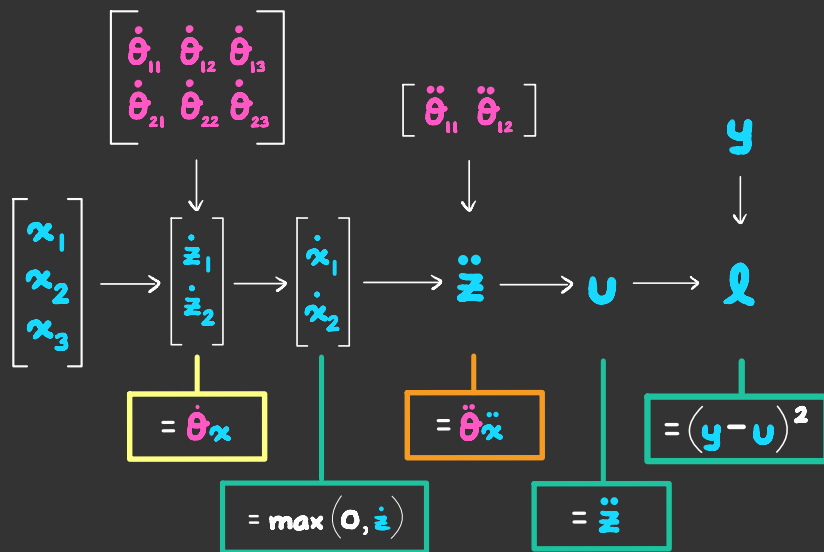
$$\frac{\partial l}{\partial \ddot{\theta}_{ij}} = (-2(y-u)) \cdot 1 \cdot \frac{\partial \ddot{z}}{\partial \ddot{\theta}_{ij}}$$



$$\frac{\partial l}{\partial \dot{\theta}_{1j}} = (-2(y-u)) \cdot 1 \cdot \ddot{\theta}_{11} \cdot \mathbb{1}(\dot{z}_1 > 0) \cdot \frac{\partial \dot{z}_1}{\partial \dot{\theta}_{1j}}$$

$$\frac{\partial l}{\partial \dot{\theta}_{2j}} = (-2(y-u)) \cdot 1 \cdot \ddot{\theta}_{12} \cdot \mathbb{1}(\dot{z}_2 > 0) \cdot \frac{\partial \dot{z}_2}{\partial \dot{\theta}_{2j}}$$

$$\frac{\partial l}{\partial \ddot{\theta}_{ij}} = (-2(y-u)) \cdot 1 \cdot \frac{\partial \ddot{z}}{\partial \ddot{\theta}_{ij}}$$



$$\frac{\partial l}{\partial \dot{\theta}_{1j}} = (-2(y-u)) \cdot 1 \cdot \ddot{\theta}_{11} \cdot \mathbb{1}(\dot{z}_1 > 0) \cdot x_j$$

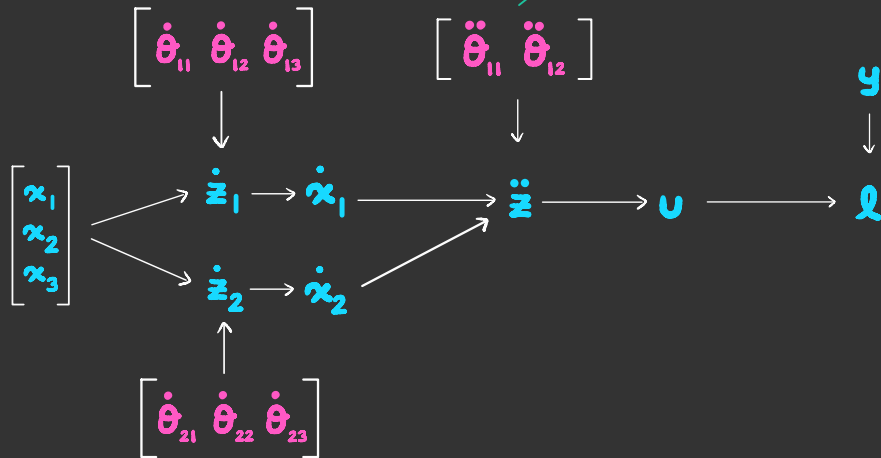
$$\frac{\partial l}{\partial \dot{\theta}_{2j}} = (-2(y-u)) \cdot 1 \cdot \ddot{\theta}_{12} \cdot \mathbb{1}(\dot{z}_2 > 0) \cdot x_j$$

$$\frac{\partial l}{\partial \ddot{\theta}_{ij}} = (-2(y-u)) \cdot 1 \cdot \dot{z}_j$$

mission accomplished!

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}_{1j}} = (-2(y-u)) \cdot \ddot{\theta}_{11} \cdot \mathbb{1}(\dot{z}_1 > 0) \cdot x_j$$

$$\frac{\partial \mathcal{L}}{\partial \ddot{\theta}_{1j}} = (-2(y-u)) \dot{x}_j$$



$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}_{2j}} = (-2(y-u)) \cdot 1 \cdot \ddot{\theta}_{12} \cdot \mathbb{1}(\dot{z}_2 > 0) \cdot x_j$$